JAN–C–719

25 JANUARY 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

CUPS, BULLET-JACKET, COPPER-PLATED STEEL
(For Small Arms Ammunition)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Frankford Arsenal
Tentative Specification
FXS–579
17 Nov. 1942

A. APPLICABLE SPECIFICATION AND DRAWINGS

A–1. *Specification.*—The following specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

JAN–S–683—Steel, Copper-Plated, Strip (For Small Arms Bullet Jackets).

Federal Specification QQ–M–151–Metals; General Specification for inspection of.

A–2. *Drawings.*—Drawings specified in the contract or purchase order shall form a part of this specification.

B. TYPE AND CONDITION

B–1. *Type.*—This specification covers one type of copper-plated steel bullet-jacket cups.

B–2. *Condition.*—Copper-plated steel cups shall be furnished in the following conditions, as specified:

Annealed.
Unannealed.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—Copper-plated steel strip from which the cups are made shall meet the requirements of National Military Establishment Specification JAN–S–683.

C–2. *Workmanship.*—The cups shall be uniform in quality, shall be free from injurious defects except as specified in paragraph E–3, and shall be suitable for the purpose for which they are intended.

C–3. *Annealed cups.*—Annealed cups shall be uniformly annealed under suitable conditions to produce the required deep drawing characteristics in the cups. The cups shall be pickled, washed and dried after annealing.

C–4. *Unannealed cups.*—Unannealed cups shall be washed and dried.

D. GENERAL REQUIREMENTS

D–1. The cups shall meet the requirements of the applicable drawings.

D–2. *Contractor's inspection.*—

D–2a. No lot of cups shall be submitted by the contractor to the Government inspector for acceptance unless, prior to such offering, the contractor has visually inspected and gaged for the defects specified in table III the required number of cups specified in table I.

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TABLE I.—*Normal lot-by-lot acceptance inspection*

Sublot size		500 to 799		800 to 1,299		1,300 to 3,199		3,200 to 7,999		8,000 to 21,999		22,000 to 109,999		110,000 to 549,999		550,000 and over		Approximate AOQL (percent defective)
First sample size		50		75		100		150		200		300		500		1,000		
Second sample size		100		150		200		300		400		600		1,000		2,000		
Acceptable quality level (percent defective)		Acceptance numbers		Acceptance numbers		Acceptance numbers		Acceptance numbers		Acceptance numbers		Acceptance numbers		Acceptance numbers		Acceptance numbers		
Major	Minor	c_1	c_2	c_1	c_2	c_1	c_2	c_1	c_2	c_1	c_2	c_1	c_2	c_1	c_2	c_1	c_2	
0.005–0.010		(*)	(*)	(*)	(*)	(*)	(*)	→	→	→	→	→	→	→	→	0	1	0.06
.011–.020	0.005–0.010	(*)	(*)	(*)	(*)	(*)	(*)	→	→	→	→	→	→	0	1	0	2	.10
.021–.030	.011–.020	(*)	(*)	(*)	(*)	→	→	→	→	→	→	0	1	0	2	0	3	.15
.031–.060	.021–.030	(*)	(*)	→	→	→	→	→	→	0	1	0	2	1	3	1	5	.25
.061–.10	.031–.060	→	→	→	→	→	→	0	1	0	2	1	2	1	4	2	7	.35
.11–.15	.061–.10	→	→	→	→	0	1	0	2	0	3	1	3	2	5	3	9	.50
.16–.25	.11–.15	→	→	0	1	0	2	1	3	1	4	2	5	3	7	4	13	.80
.26–.50	.16–.25	0	2	1	2	1	3	2	4	2	7	3	8	4	13	7	23	1.0
.51–1.0	.26–.50	1	3	2	4	2	6	3	9	4	11	5	13	7	23	14	40	2.0
1.1–2.0	.51–1.0	2	4	3	5	3	8	5	13	6	16	8	25	14	40	←	←	3.0
2–1–3.0	1.1–2.0	3	5	4	8	5	11	7	18	9	24	12	35	←	←	←	←	4.0
3.1–4.0	2.1–3.0	3	9	5	11	6	16	9	23	11	32	16	47	←	←	←	←	5.0
4.1–5.0	3.1–4.0	4	10	6	14	8	19	11	28	14	37	←	←	←	←	←	←	6.0
	4.1–5.0	5	12	7	17	10	23	14	33	18	44	←	←	←	←	←	←	8.0

*Table not applicable in this region.

→Use sample size in first columns to right in which acceptance numbers are shown for acceptable quality level involved, except as indicated under note 4 below.

←Use sample size in first columns to left in which acceptance numbers are shown for acceptable quality level involved.

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PROCEDURE:

1. For major defects.
   a. Select first sample of size indicated in table I for sublot size involved.
   b. Determine in first sample the number of articles, d_1, which contain major defects.
      (1) If d_1 does not exceed the c_1 indicated for the sample size and acceptable quality level, major, involved; pass sublot for majors.
      (2) If d_1 exceeds the corresponding c_2. Return sublot to contractor.
      (3) If d_1 exceeds c_1, but does not exceed c_2:
         (a) Select second sample of size indicated in table I.
         (b) Determine in second sample the number of articles, d_2, which contain major defects.
            1. If d_1+d_2 does not exceed c_2. Pass sublot for majors.
            2. If d_1+d_2 exceeds c_2: Return sublot to contractor.
2. For minor defects:
   Carry out above procedure with "minor" substituted everywhere for "major," using same sample wherever feasible.
3. Return to contractor all defective articles observed in any of the above inspections.

NOTES:

4. Above procedure is satisfactory when process average is equal to or better than acceptable quality level. If process average is poorer than this, use acceptance numbers that correspond with the AOQL value which is equal to or next better than the acceptable quality level.
5. AOQL, the average outgoing quality limit, is the poorest average quality accepted if all sublots returned to contractor are inspected 100 percent and accepted after removal of all defective articles.

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TABLE II.—*Reduced lot-by-lot acceptance inspection*

Sublot size		500 to 799			800 to 1,299			1,300 to 3,199			3,200 to 7,999			8,000 to 21,999			22,000 to 109,999			110,000 to 549,999			550,000 and over		
First sample size		10			15			20			30			40			60			100			200		
Second sample size		140			210			280			420			560			840			1,400			2,800		
Acceptable quality level (percent defective)		Acceptance numbers			Acceptance numbers			Acceptance numbers			Acceptance numbers			Acceptance numbers			Acceptance numbers			Acceptance numbers			Acceptance numbers		
Major	Minor	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e	c_1	c_2	c_e
0.005–0.010		(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	→	→	→	→	→	→	→	→	→	→	→	→	0	1	0
.011– .020	0.005–0.010	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	→	→	→	→	→	→	→	→	→	→	→	→	0	2	0
.021– .030	.011– .020	(*)	(*)	(*)	(*)	(*)	(*)	→	→	→	→	→	→	→	→	→	0	1	0	0	2	0	0	3	0
.031– .060	.021– .030	(*)	(*)	(*)	→	→	→	→	→	→	→	→	→	0	1	0	0	2	0	0	3	1	0	5	1
.061– .10	.031– .060	→	→	→	→	→	→	→	→	→	0	1	0	0	2	0	0	2	1	0	4	1	1	7	2
.11– .15	.061– .10	→	→	→	→	→	→	0	1	0	0	2	0	0	3	0	0	3	1	1	5	2	1	9	3
.16– .25	.11– .15	→	→	→	0	1	0	0	2	0	0	3	1	0	4	1	1	5	2	1	7	3	1	13	4
.26– .50	.16– .25	0	2	0	0	2	1	0	3	1	1	4	2	1	7	2	1	8	3	1	13	4	2	23	7
.51–1.0	.26– .50	0	3	1	1	4	2	1	6	2	1	9	3	1	11	4	2	13	5	2	23	7	4	40	14
1.1–2.0	.51–1.0	1	4	2	1	5	3	1	8	3	2	13	5	2	16	6	3	25	8	4	40	14	←	←	←
2.1–3.0	1.1–2.0	1	5	3	1	8	4	2	11	5	2	18	7	3	24	9	4	35	12	←	←	←	←	←	←
3.1–4.0	2.1–3.0	1	9	3	2	11	5	2	16	6	3	23	9	3	32	11	5	47	16	←	←	←	←	←	←
4.1–5.0	3.1–4.0	1	10	4	2	14	6	3	19	8	3	28	11	4	37	14	←	←	←	←	←	←	←	←	←
	4.1–5.0	2	12	5	3	17	7	4	23	10	4	33	14	5	44	18	←	←	←	←	←	←	←	←	←

*Table not applicable in this region.
→ Use sample size in first columns to right in which acceptance numbers are shown for acceptable quality level involved.
← Use sample size in first columns to left in which acceptance numbers are shown for acceptable quality level involved.

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CONDITIONS UNDER WHICH REDUCED INSPECTION (TABLE II) MAY BE STARTED FOR ANY PARTICULAR CLASS OF DEFECTS:

1. Production has been continuous for the immediately preceding 2 weeks, and for not less than 20 sublots; and
2. The last 20 sublots have been inspected under table I, and none have been returned to the contractor because of failure to meet the require ments of table I for the class of defects involved; and
3. The percent defective, computed from the combined first samples from these 20 sublots and all other sublots inspected during the last 2-week period, is less than the lower boundary of the acceptable quality range being used (left-hand columns of table I).

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PROCEDURE UNDER REDUCED INSPECTION (TABLE II):

4. Select first sample, pass sublot, return sublot to contractor, select second sample, etc., as in procedure for table I, but using sample sizes and acceptance numbers, c_1 and c_2 of table II.
5. In addition to action on individual sublots under paragraph 4, the first five such sublots inspected must be considered as a group, the second five as a group, etc.
   a. As the second, third, etc., sublot in each group is inspected, cumulate the number of defectives (articles which contain defect of the class involved) in first samples from that group.
   b. If at any time during the inspection of successive sublots in a group, the cumulated number of defectives, d_e, in the first samples so far inspected exceeds the c_e indicated in table II, select second sample from the current sublot and determine the number of defectives, d_2, in this second sample. If d_1+d_2 for the current sublot exceeds c_2, return that sublot to contractor; otherwise, pass sublot for the class of defects involved. (d_1 is number of defectivesin first sample from current sublot.)
   c. Whenever a second sample is inspected from any sublot, the next sublot is to become the first in a new group of five successive sublots.
6. Return to contractor all defective articles observed in any of the above inspections.

CONDITIONS UNDER WHICH NORMAL INSPECTION (TABLE I) MUST BE RESUMED FOR THE CLASS OF DEFECTS INVOLVED:

7. Each succeeding sublot shall be inspected under table I.
   a. If a sublot is returned to contractor under paragraph 4 or 5b; or
   b. If d_e (determined under paragraph 5a) plus d_2 (determined under paragraph 4 or 5b) exceeds the c_2 indicated in table II.
8. Inspection under table I shall then continue until the conditions of paragraphs 1, 2, and 3 are again met.

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D–2b. A lot shall be submitted only if the number of defects found by visual inspection and gaging are within the limits specified in table I at an acceptable quality level equal to or better than that specified in paragraph E–3.

D–2c. At the time of submission of a lot, the contractor shall furnish the inspector a report, in duplicate, showing the results of the visual inspection and gaging made by him in accordance with the requirements specified in paragraphs D–2a and D–2b.

E. DETAIL REQUIREMENTS

E–1. *Hardness of annealed cups.*—When cups are specified in the annealed condition, the hardness of any cup shall not exceed Rockwell B60.

E–2. The cups shall process satisfactorily into bullet jackets.

E–3. *Visual and dimensional defects.*—Visual and dimensional defects shall be as defined in table III.

TABLE III.—*Visual and dimensional defects*

Visual		Dimensional
Major	*Minor*	*Major*
Defective plating	Scratches	Base thickness, thick or thin
Improperly formed or mutilated	Oil, grease, dirt	Wall thickness, thick or thin [3]
Laminations [1]	Oxidation [2]	Outside diameter, large or small
Scrap	Dents or bent edges	Weight, light or heavy
		Excessive wall height variation

[1] In order to determine whether or not apparent laminations shall be counted as defects, cups shall be sectioned, polished, etched, and examined with a microscope. If a break extends more than 0.020 inch below the mouth edge, the cups shall be called laminated. Seams and cracks should be examined after sectioning. If the seams and cracks extend into the body of the metal, the cups shall be considered defective.

[2] Oxidation includes discoloration due to the failure in pickling and rinsing to remove the oxides formed in annealing.

[3] It should be determined that every point on the specified periphery is within the wall thickness limits.

E–3a. *Acceptable quality levels.*—The acceptable quality levels (percent defective) of the cups shall not exceed the percents shown in table IV.

TABLE IV.—*Acceptable quality levels (percent defective)*

Visual		Dimensional
Major	Minor	Major
Percent 0.25	*Percent* 1.0	*Percent* 2.0

E–3b. The number of defective cups in each classification of visual inspection and gaging defects shall not exceed the number specified in table I or II, whichever is used, for the specified acceptable quality levels.

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F. METHOD OF SAMPLING, INSPECTION, AND TESTS

F–1. The inspection and tests, unless otherwise specified, shall be made at the place of manufacture.

F–2. *Inspection gages.*—The gaging of any sample cups required by this specification, except that specified in paragraph D–2, shall be made by gages belonging to the Government and in charge of the inspector.

F–3.—*Size of lot.*—A lot shall consist only of material from one heat of steel, of the same size cups, of the same condition and shall not exceed 2,000,000 pieces or carload capacity, whichever is smaller.

F–4. *Sampling.*—The inspector shall select from the lot, samples in accordance with the requirements specified in tables I and II. The samples shall be selected in equal quantities from at least one-fourth of the containers holding the cups in the lot.

F–5. *Visual inspection and gaging.*—Each cup of the first inspection sample shall be visually inspected and gaged for the defects specified in table III. The second sample shall be inspected when required by the procedure specified in table I or II, whichever is used.

F–6. *Hardness test.*—A minimum of 25 cups taken from the sample for visual inspection and gaging shall be used for hardness determinations. Each cup of the sample shall be sectioned longitudinally. The copper plating shall be removed and any surface defect removed by grinding or polishing. The specimen shall then be mounted on a flat anvil with the concave surface upward and readings taken from near the center of the sidewall.

F–7. *Work test.*

F–7a. When required by the contracting officer, a minimum random sample or 4,000 cups shall be subjected to a work test which shall comprise all the operations used in the manufacture of the appropriate jacketed bullets.

F–7b. The results of the work test specified in National Military Establishments Specification JAN–S–683 on the strip from which the lot of cups under acceptance is made, at the option of the contracting officer, may be taken as the work test specified in paragraph F–7a.

F–8. *Retests.*

F–8a. Retests for tests other than work tests may be allowed only upon the written request of the contractor, addressed to and approved by the contracting officer, and shall be in accordance with the requirements specified in paragraphs F–8c and F–8d.

F–8b. No retest shall be allowed on a lot if it fails in the work test.

F–8c. *Visual inspection and gaging.*—A lot of cups which has been presented and rejected because of the failure to meet the acceptable qualities specified for visual and dimensional defects may be resubmitted. However, prior to such resubmission, the contractor shall remove from the lot all defective cups. Retest sample sizes and acceptance numbers shall be those specified in table I.

F–8d. *Hardness.*—If a retest for hardness is made, a sample of 50 cups shall be used. All cups in the retest sample shall meet the hardness requirements.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*

G–1a. The cups shall be packed in wooden barrels, drums, or boxes, or moisture resistant fiberboard boxes approved by the contracting officer, having the following weights for each type of container. All

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wooden containers shall be paper lined. Packing in sawdust shall not be permitted.

(1) Wooden barrels—total weight with contained cups, not over 1,000 pounds.

(2) Wooden drums—total weight with contained cups, not over 650 pounds.

(3) Wooden boxes or moisture-resistant fiberboard boxes—total weight with contained cups, not over 150 pounds.

G–1b. All material shall be properly segregated by lots when prepared for shipment.

G–1c. All material shall be protected against corrosion in transit.

G–1d. Material shall be packed in such manner as to insure acceptance by common or other carriers for safe transportation at lowest rate to the point of delivery.

G–2. *Marking.*—Each container shall be legibly and indelibly marked with the name of the material, condition, size, lot number, name of the contractor, the name of the steel manufacturer, the heat number of the steel, and the contract or purchase order number.

H. NOTES

H–1. The cups covered by this specification are intended for use in the manufacture of copper plated steel bullet jackets for small arms ammunition.

H–2. The following publications specify the methods used by the Government inspector for visually inspecting and gaging the cups under test:

Ordnance Gage Manual for Small Arms Ammunition Components.

Visual Inspection Standards for Cups Used in Small Arms Ammunition Manufacture.

Standard Inspection Procedures, Issue 2, dated 3 February 1944.

H–3. Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

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